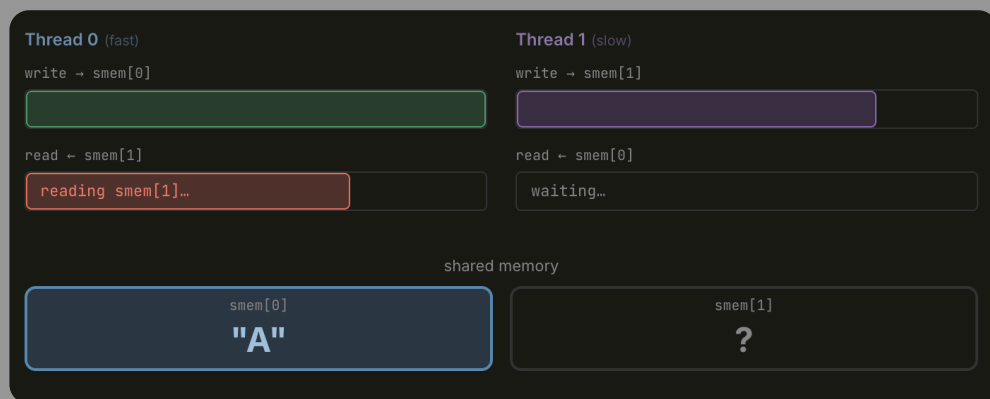


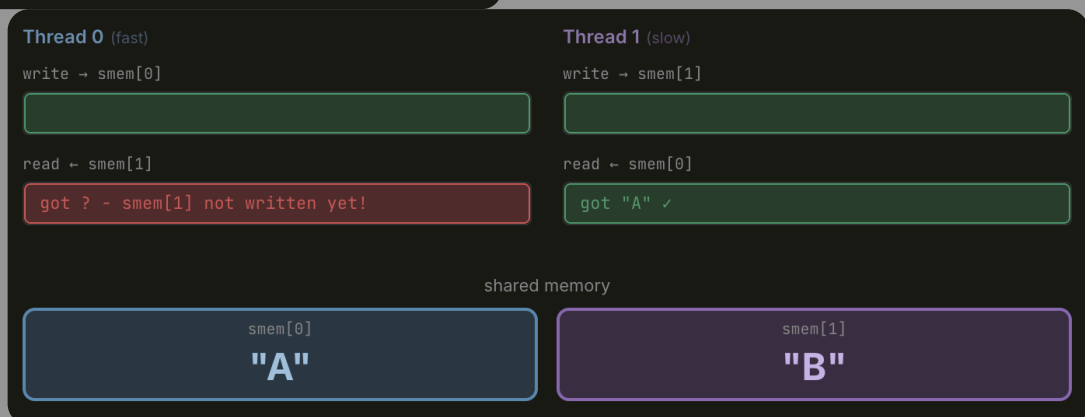
→ The Problem

⇒ Race Conditions



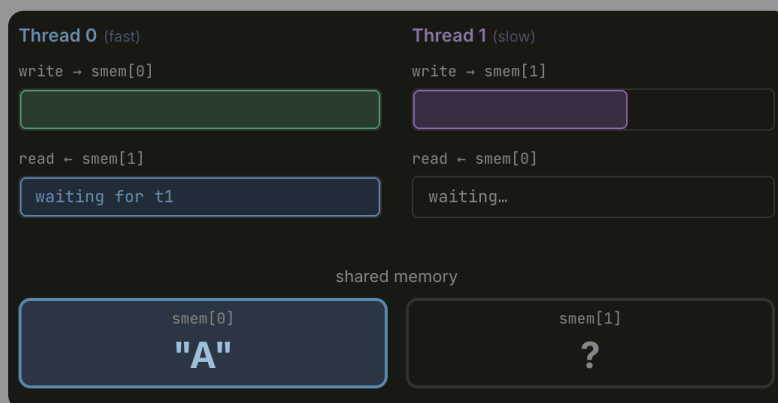
> fast thread can read before a slow thread has finished writing

Causing a corrupted output the kernel never reports

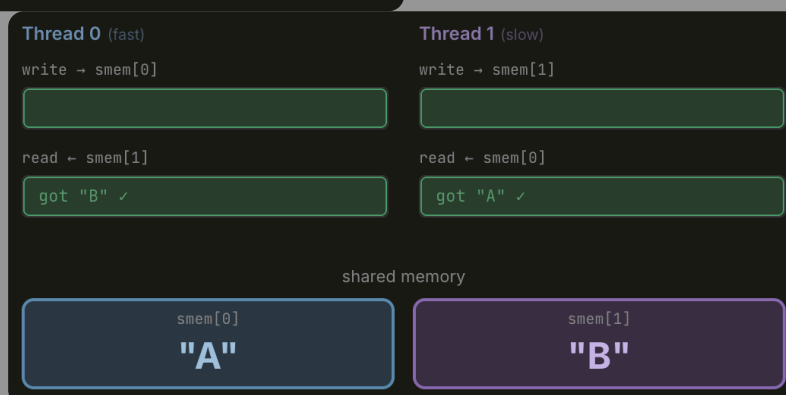


→ The fix

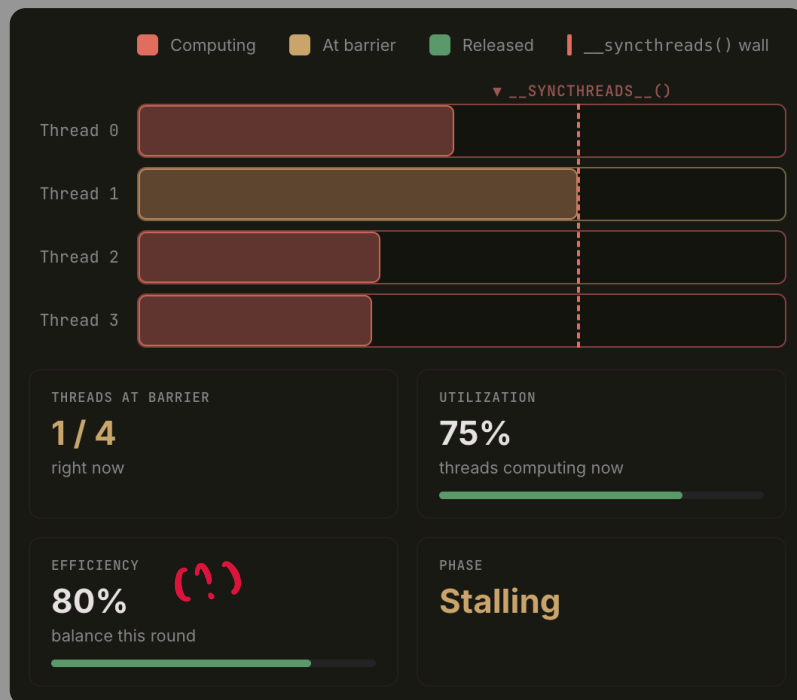
⇒ Read waits for write



> Thread 0 waits for thread 1's write to finish before reading from smem[1], eliminating the race cond.

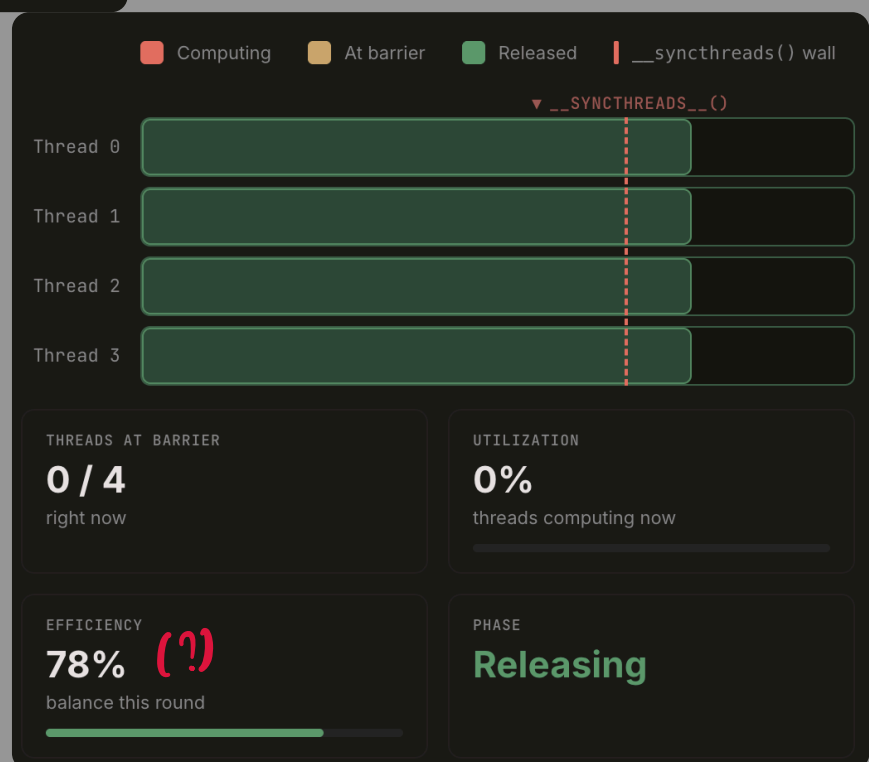


⇒ Block Barrier



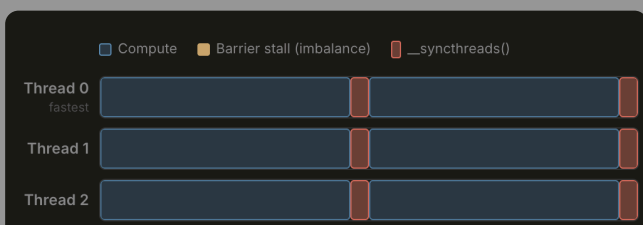
> fast threads stall at the barrier,
locking the entire block

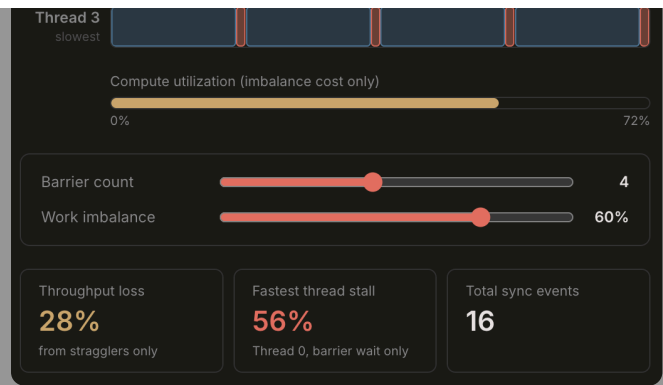
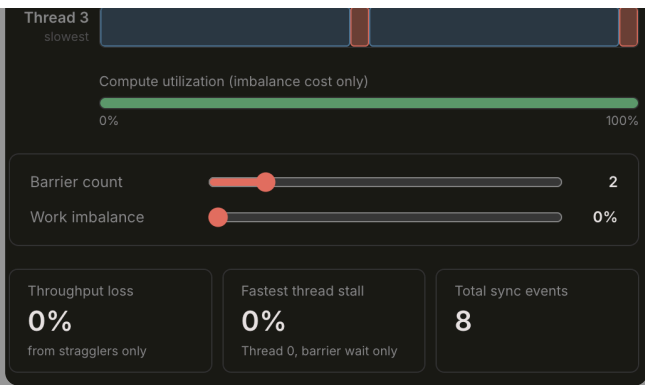
until the slowest
thread arrives for all
to continue



→ The trade-off

⇒ why barriers cost





➤ The cost is idle threads waiting for the slowest one and every extra barrier repeats that wait.